

EDUCATION

Purdue University <i>B.Sc. Computer Engineering Technology, Minor in Information Technology</i>	West Lafayette, IN 08/2023 – 05/2027
- Researching with SoCET: Developing our custom trace-scheduling compiler for the in-house VLIW AI Accelerator SoC "Atalla". - Completed all embedded programming coursework in degree path and audited advanced CS and ECE courses beyond curriculum requirements including Data Structures & Algorithms and a graduate-level Compiler Design course.	

PROJECTS

Modernized Superheterodyne FM Radio Receiver	04/2025
- Engineered embedded C firmware around an Arduino Mega to digitally tune a fully analog superheterodyne FM receiver using an I2C 12-bit DAC, ADC RSSI feedback, EEPROM station storage, and LCD interface. - Implemented a DSP-assisted channel scan algorithm that computed RF noise floor, identified RSSI peaks, and auto-locked onto stations through progressively finer frequency sweeps across the 88–108 MHz FM band. - Characterized VCO behavior using quartic regression across 56 measured data points, achieving an R^2 of 0.9987 and enabling highly accurate digital frequency estimation and automatic station calibration.	
Embedded Sensor Control + BLE Interface	04/2026
- Built a closed-loop embedded control system using a Raspberry Pi Pico, MOSFET/BJT analog driver circuitry, and MicroPython PID firmware to regulate thermal output in real time. - Integrated Bluetooth Low Energy (BLE) serial communication for wireless setpoint configuration and live telemetry streaming, enabling rapid hardware validation and debugging during bench testing. - Performed bring-up and verification of mixed-signal circuitry including transistor driver stages and sensor feedback loops.	
Industrial Control Systems + PLC Development	09/2025
- Built industrial automation systems using IEC 61131-3 standards across Phoenix Contact and Allen-Bradley PLC platforms. - Designed and simulated operator HMIs prior to deployment using PLCnext Engineer, FactoryTalk Design Workbench, and Studio5000 to validate control logic and reduce integration errors. - Applied real-time systems concepts through FreeRTOS task scheduling labs and concurrent FPGA/HDL simulations in Quartus to strengthen embedded debugging and hardware-software integration skills.	
Engine Monitoring Android App	02/2024
- Built an Android driver-assistance app that streamed OBD-II telemetry over MQTT and computed real-time drivetrain insights from engine sensor data. - Derived current transmission gear mathematically from engine RPM, vehicle speed, and final drive ratios when no direct PID was available from the ECU.	

EXPERIENCE

Purdue Electric Racing <i>Embedded Systems Team Member</i>	West Lafayette, IN 09/2024 – 12/2025
- Architected a register-level PWM HAL for STM32 vehicle subsystems, reducing fan-control implementation from ~40 lines to 2 while improving firmware reuse and maintainability. - Validated CAN, PWM, and sensor communication reliability using oscilloscopes, logic analyzers, and CANable adapters on safety-critical automotive systems. - Refined vehicle CAN topology by isolating high-throughput motor traffic onto a dedicated network to improve bus stability.	
Wistron <i>NVIDIA Server Repair Technician</i>	Dallas, TX 05/2024 – 08/2024
- Diagnosed HGX 8-GPU server failures involving FPGAs, power delivery, and signal integrity using proprietary validation tools. - Created Python automation tooling that removed thousands of lines of repetitive database entry, saving 80+ labor hours weekly. - Executed hardware bring-up, repair verification, and subsystem validation under strict throughput and quality targets.	
Jawa.gg <i>Verified Seller – PCs & Components</i>	Remote 10/2021 – 08/2023
- Ranked Top 10 platform-wide in seller performance and recognized by CEO Amanda Stefan for contributions to marketplace growth. - Reverse-engineered board-level failures using schematics and BoardView files; completed BGA rework and precision micro-soldering.	

SKILLS

Languages: C, Python, Java, Rust, MicroPython, Bash, Structured Text
Embedded/Firmware: Embedded C, FreeRTOS, STM32, Arduino, Raspberry Pi Pico, ESP8266, HAL, Board Bring-Up
Interfaces/Protocols: I2C, SPI, UART, USB, BLE, MQTT, GPIO, PWM, ADC, DAC, OBD-II
Development Tools: Git/GitHub, PlatformIO, STM32Cube, CMake, Multisim, PLCnext Engineer, Studio5000, Linux
Hardware/Testing: PCB Design, Bench Testing, Oscilloscopes, Logic Analyzers, Embedded Debugging, Signal Analysis, Soldering
Concepts: Agile Development, Continuous Integration, Software Validation, Embedded Operating Systems, Functional Testing, Controller Integration, Troubleshooting, Software Documentation

RELEVANT COURSEWORK

Advanced Embedded Digital Systems Wireless Communications Concurrent Digital Systems Industrial Controls
DAQ & Systems Control Electronics Prototyping & Development DC & Pulse Electronics Data Structures & Algorithms